

Some partial solutions and/or hints are given below. Watch out for typographical errors and misprints. If you catch an error, please let me know.

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1. Dirac notation (bra-ket notation).

- (a) [1 pt.] If $|\phi\rangle = c|\chi\rangle$, where c is a complex number, express $\langle\phi|$ in terms of $\langle\chi|$.

Solution/Hints $\rightarrow$

$$|\phi\rangle = c|\chi\rangle \quad \implies \quad \langle\phi| = c^*\langle\chi|$$

- (b) [1 pt.] If $|\phi\rangle = \hat{A}|\chi\rangle$, where $\hat{A}$ is an operator (not necessarily hermitian), express $\langle\phi|$ in terms of $\hat{A}$ and $\langle\chi|$.

Solution/Hints $\rightarrow$

$$|\phi\rangle = \hat{A}|\chi\rangle \quad \implies \quad \langle\phi| = \langle\chi|\hat{A}^\dagger$$

Note: this is NOT written as $\hat{A}^\dagger\langle\chi|$. An operator acting on a dual/bra vector is written to the right of the bra, not to the left.

Alternately, the notation

$$\langle\phi|\hat{A}^\dagger = \langle(\hat{A}\phi)|$$

could also be used, i.e., $\langle\phi|\hat{A}^\dagger$ is the dual vector of $\hat{A}|\phi\rangle$.

- (c) [SELF] Is $\langle\chi|\phi\rangle$ a number, an operator, or a ket vector, or a bra vector? What about $|\chi\rangle\langle\phi|$?

Solution/Hints $\rightarrow$

$\langle\chi|\phi\rangle$ is a number. It is the inner product of the vectors $|\chi\rangle$ and $|\phi\rangle$. You can think of it as a row vector multiplying a column vector, which would result in a number.

$|\chi\rangle\langle\phi|$ is an operator. In matrix/vector representation, you can think of it as a column vector multiplying a row vector, which would result in a matrix. Matrices represent operators.

- (d) [1 pt.] If $|\phi\rangle$ is not normalized, what would you multiply it by, in order to obtain a normalized state vector? Are you multiplying by a number, an operator, or another ket?

Solution/Hints $\rightarrow$

We would multiply by the factor

$$\lambda = \frac{1}{\sqrt{\langle\phi|\phi\rangle}}$$

This is a NUMBER. The resulting state vector $|\psi\rangle = \lambda |\phi\rangle = \frac{1}{\sqrt{\langle\phi|\phi\rangle}} |\phi\rangle$ is normalized:

$$\langle\psi|\psi\rangle = \lambda^2 \langle\phi|\phi\rangle = \left(\frac{1}{\sqrt{\langle\phi|\phi\rangle}}\right)^2 \langle\phi|\phi\rangle = \frac{1}{\langle\phi|\phi\rangle} \langle\phi|\phi\rangle = 1$$

$\sqrt{\langle\phi|\phi\rangle}$ is called the NORM of ϕ .

To be more general, we could include a multiplicative phase factor with λ , which would keep the norm unchanged:

$$\lambda = \frac{e^{ir}}{\sqrt{\langle\phi|\phi\rangle}} \quad \begin{cases} r \text{ is a real number, so that} \\ |e^{ir}|^2 = 1 \text{ and } |\lambda|^2 = \lambda^2 \end{cases}$$

- (e) [1 pt.] If $|\chi\rangle$ is normalized and $|\phi\rangle = \gamma |\chi\rangle$, where γ is a complex number, find the value of $\langle\chi|\phi\rangle$.

Solution/Hints $\rightarrow$

$$\langle\chi|\phi\rangle = \langle\chi|\gamma|\chi\rangle = \gamma\langle\chi|\chi\rangle = \gamma$$

- (f) [1 pt.] If $\langle\chi|\phi\rangle = \beta$, then what is $\langle\phi|\chi\rangle$?

Solution/Hints $\rightarrow$

$$\langle\phi|\chi\rangle = \langle\chi|\phi\rangle^* = \beta^*$$

2. Basis vectors.

A physical system is described by a three-dimensional Hilbert space. Any state vector can be expressed as a linear combination of the three state vectors $|\phi_1\rangle$, $|\phi_2\rangle$, $|\phi_3\rangle$. We say that the set $\{|\phi_j\rangle\}$ is a basis set (a set of basis vectors) for this space.

We choose our three basis vectors to be **orthonormal**.

- (a) [1 pt.] Find out (look up) what orthonormal means. Our basis set $\{|\phi_j\rangle\}$ is orthonormal: Express what this means in terms of inner products. In particular, what is $\langle\phi_1|\phi_1\rangle$, and what is $\langle\phi_1|\phi_2\rangle$?

Solution/Hints $\rightarrow$

Orthonormal means each of the basis vectors is normalized and is orthogonal to all the other basis vectors. This can be written as

$$\langle\phi_i|\phi_j\rangle = \delta_{ij} \quad \begin{cases} \text{where } \delta_{ij} \text{ is the Kronecker delta:} \\ = 0 \text{ when } i \neq j \text{ and } = 1 \text{ when } i = j \end{cases}$$

In particular, $\langle\phi_1|\phi_1\rangle = 1$ and $\langle\phi_1|\phi_2\rangle = 0$.

- (b) [2 pts.] Consider the normalized state

$$|W\rangle = \gamma |\phi_1\rangle + \gamma |\phi_2\rangle + \gamma |\phi_3\rangle.$$

Determine γ . Assume γ to be purely imaginary.

Solution/Hints $\rightarrow$

If the vector $|S\rangle$ is expressed in terms of the basis vectors as

$$|S\rangle = \sum_j s_j |\phi_j\rangle$$

then using orthonormality of the set $\{|\phi_j\rangle\}$ it's easy to show

$$\langle S|S\rangle = \sum_j s_j^* s_j = \sum_j |s_j|^2$$

Please make sure that you can work out how this follows from orthonormality of the basis vectors.

Thus the norm of the vector $|W\rangle$ is

$$\langle W|W\rangle = |\gamma|^2 + |\gamma|^2 + |\gamma|^2 = 3|\gamma|^2$$

Since the state $|W\rangle$ is normalized, this has to be unity, i.e., $3|\gamma|^2 = 1$. Since γ is purely imaginary, this means

$$\gamma = \frac{i}{\sqrt{3}} \quad \text{or} \quad \gamma = -\frac{i}{\sqrt{3}}$$

We will use the first form below when $|W\rangle$ is referred to.

(c) [**2 pts.**] Consider the states

$$|\psi\rangle = \frac{1}{\sqrt{3}}|\phi_1\rangle - \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{i}{\sqrt{3}}|\phi_3\rangle; \quad |\theta\rangle = \frac{i}{2}|\phi_1\rangle + \frac{\sqrt{3}}{2}|\phi_3\rangle$$

Show that these two states are normalized.

Solution/Hints $\rightarrow$

Norm of $|\psi\rangle$ is

$$\langle\psi|\psi\rangle = \left|\frac{1}{\sqrt{3}}\right|^2 + \left|-\frac{1}{\sqrt{3}}\right|^2 + \left|\frac{i}{\sqrt{3}}\right|^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

and similarly for $|\theta\rangle$.

(d) [**1 pt.**] Express $\langle\psi|$ in terms of the bra vectors corresponding to the basis vectors, i.e., in terms of the set $\{\langle\phi_j|\}$.

Solution/Hints $\rightarrow$

$$\langle\psi| = \frac{1}{\sqrt{3}}\langle\phi_1| - \frac{1}{\sqrt{3}}\langle\phi_2| - \frac{i}{\sqrt{3}}\langle\phi_3|$$

Note that the coefficients are complex conjugated.

- (e) [**2 pts.**] Calculate $\langle \psi | \theta \rangle$ and $\langle \theta | \psi \rangle$.

Solution/Hints $\rightarrow$

If the vectors $|S\rangle$ and $|T\rangle$ are expressed in terms of the basis vectors as

$$|S\rangle = \sum_j s_j |\phi_j\rangle \quad \text{and} \quad |T\rangle = \sum_j t_j |\phi_j\rangle$$

then using orthonormality of the set $\{|\phi_j\rangle\}$ it's easy to show

$$\langle S | T \rangle = \sum_j s_j^* t_j \quad \text{and} \quad \langle T | S \rangle = \sum_j t_j^* s_j$$

Please make sure you are comfortable using this and can work out how this follows from orthonormality.

It follows:

$$\begin{aligned} \langle \psi | \theta \rangle &= \left(\frac{1}{\sqrt{3}} \right) \left(\frac{i}{2} \right) + \left(-\frac{1}{\sqrt{3}} \right) (0) + \left(-\frac{i}{\sqrt{3}} \right) \left(\frac{\sqrt{3}}{2} \right) \\ &= \frac{i}{2\sqrt{3}} - \frac{i}{2} = \frac{i}{2} \left(\frac{1}{\sqrt{3}} - 1 \right) \\ \langle \theta | \psi \rangle &= -\frac{i}{2} \left(\frac{1}{\sqrt{3}} - 1 \right) \end{aligned}$$

- (f) [**3 pts.**] Let's define the operator $\hat{A} = |\theta\rangle\langle\psi|$. Write out the operator in terms of the basis vectors and their duals, i.e., in terms of operators $|\phi_i\rangle\langle\phi_j|$.

Solution/Hints $\rightarrow$

$$\begin{aligned} \hat{A} = |\theta\rangle\langle\psi| &= \left(\frac{i}{2} |\phi_1\rangle + \frac{\sqrt{3}}{2} |\phi_3\rangle \right) \left(\frac{1}{\sqrt{3}} \langle\phi_1| - \frac{1}{\sqrt{3}} \langle\phi_2| - \frac{i}{\sqrt{3}} \langle\phi_3| \right) \\ &= \frac{i}{2\sqrt{3}} |\phi_1\rangle\langle\phi_1| - \frac{i}{2\sqrt{3}} |\phi_1\rangle\langle\phi_2| + \frac{1}{2\sqrt{3}} |\phi_1\rangle\langle\phi_3| \\ &\quad + \frac{1}{2} |\phi_3\rangle\langle\phi_1| - \frac{1}{2} |\phi_3\rangle\langle\phi_2| - \frac{i}{2} |\phi_3\rangle\langle\phi_3| \end{aligned}$$

- (g) [1 pt.] Find the state obtained by operating with $\hat{A}$ on the state $|\phi_1\rangle$.

Solution/Hints $\rightarrow$

Since $\langle\phi_m|\phi_n\rangle = \delta_{mn}$, the only terms in $\hat{A}$ which contribute when applied on $|\phi_1\rangle$ are those of the form $|\phi_j\rangle\langle\phi_1|$. In other words,

$$\begin{aligned} \left(|\phi_j\rangle\langle\phi_1|\right)|\phi_1\rangle &= |\phi_j\rangle\langle\phi_1|\phi_1\rangle = |\phi_j\rangle \\ \left(|\phi_j\rangle\langle\phi_2|\right)|\phi_1\rangle &= |\phi_j\rangle\langle\phi_2|\phi_1\rangle = 0 \\ \left(|\phi_j\rangle\langle\phi_3|\right)|\phi_1\rangle &= |\phi_j\rangle\langle\phi_3|\phi_1\rangle = 0 \end{aligned}$$

Hence, only the first and fourth terms in the above expression for $\hat{A}$ will contribute to $\hat{A}|\phi_1\rangle$. Thus

$$\hat{A}|\phi_1\rangle = \frac{i}{2\sqrt{3}}|\phi_1\rangle + \frac{1}{2}|\phi_3\rangle$$

- (h) [2 pts.] Find the state obtained by operating with $\hat{A}$ on the state $|W\rangle$, i.e., express $\hat{A}|W\rangle$ in terms of the basis vectors.

Solution/Hints $\rightarrow$

$$\begin{aligned} \hat{A}|W\rangle &= \left(\frac{i}{2\sqrt{3}}|\phi_1\rangle\langle\phi_1| - \frac{i}{2\sqrt{3}}|\phi_1\rangle\langle\phi_2| + \frac{1}{2\sqrt{3}}|\phi_1\rangle\langle\phi_3| \right. \\ &\quad \left. + \frac{1}{2}|\phi_3\rangle\langle\phi_1| - \frac{1}{2}|\phi_3\rangle\langle\phi_2| - \frac{i}{2}|\phi_3\rangle\langle\phi_3| \right) \left(\frac{i}{\sqrt{3}}|\phi_1\rangle + \frac{i}{\sqrt{3}}|\phi_2\rangle + \frac{i}{\sqrt{3}}|\phi_3\rangle \right) \end{aligned}$$

Each term of $\hat{A}$, when applied to $|W\rangle$, gives exactly one nonzero term due to orthonormality of the basis vectors. Hence:

$$\begin{aligned} \hat{A}|W\rangle &= -\frac{1}{6}|\phi_1\rangle + \frac{1}{6}|\phi_1\rangle + \frac{i}{6}|\phi_1\rangle + \frac{i}{2\sqrt{3}}|\phi_3\rangle - \frac{i}{2\sqrt{3}}|\phi_3\rangle + \frac{1}{2\sqrt{3}}|\phi_3\rangle \\ &= \frac{i}{6}|\phi_1\rangle + \frac{1}{2\sqrt{3}}|\phi_3\rangle \end{aligned}$$

- (i) [**2 pts.**] Let's introduce the shorthand notation $A_{ij} = \langle \phi_i | \hat{A} | \phi_j \rangle$. Find the values of the nine quantities A_{ij} . These are generally known as 'matrix elements' of the operator $\hat{A}$.

Solution/Hints $\rightarrow$

I will write the nine quantities as elements of a matrix.

$$\begin{pmatrix} \frac{i}{2\sqrt{3}} & -\frac{i}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} \\ 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{i}{2} \end{pmatrix}$$

3. Vector/matrix notation. (Building on previous problem)

We are using the set $\{|\phi_j\rangle\}$ as a basis set. This can be expressed in vector notation as

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad |\phi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

The size of the vectors is the number of orthonormal basis vectors required to represent all states, which is also the dimension of the Hilbert space. (In this case: three.)

- (a) [**1 pt.**] Write down the states $|W\rangle$, $|\psi\rangle$ and $|\theta\rangle$ (from previous problem) as vectors.

Solution/Hints $\rightarrow$

$$|W\rangle = \gamma |\phi_1\rangle + \gamma |\phi_2\rangle + \gamma |\phi_3\rangle = \gamma \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \gamma \\ \gamma \\ \gamma \end{pmatrix}$$

Similarly

$$|\psi\rangle = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{i}{\sqrt{3}} \end{pmatrix} \quad |\theta\rangle = \begin{pmatrix} \frac{i}{2} \\ 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix}$$

- (b) [**2 pts.**] Write down the row vectors which represent the dual (bra) vectors $\langle\psi|$ and $\langle\theta|$.

Solution/Hints $\rightarrow$

In vector notation, the dual vectors are simply the hermitian conjugates (adjoints) of the vectors. Hence bra vectors are represented by hermitian conjugates column vectors, and hence are row vectors.

$$\langle\psi| = \left(\frac{1}{\sqrt{3}} \quad -\frac{1}{\sqrt{3}} \quad -\frac{i}{\sqrt{3}} \right) \quad \langle\theta| = \left(-\frac{i}{2} \quad 0 \quad \frac{\sqrt{3}}{2} \right)$$

Note the complex conjugates!!

- (c) [**1 pt.**] Using these representations, calculate $\langle\psi|\theta\rangle$ through matrix multiplication between a row vector and a column vector. (Check with the value that you calculated previously using bra-ket notation.)

Solution/Hints $\rightarrow$

$$\begin{aligned} \langle\psi|\theta\rangle &= \left(\frac{1}{\sqrt{3}} \quad -\frac{1}{\sqrt{3}} \quad -\frac{i}{\sqrt{3}} \right) \begin{pmatrix} \frac{i}{2} \\ 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} \\ &= \left(\frac{1}{\sqrt{3}} \right) \left(\frac{i}{2} \right) + \left(-\frac{1}{\sqrt{3}} \right) (0) + \left(-\frac{i}{\sqrt{3}} \right) \left(\frac{\sqrt{3}}{2} \right) \\ &= \frac{i}{2\sqrt{3}} - \frac{i}{2} = \frac{i}{2} \left(\frac{1}{\sqrt{3}} - 1 \right) \end{aligned}$$

- (d) [**3 pts.**] Construct the matrix representing the operator $\hat{A} = |\theta\rangle\langle\psi|$, by multiplying the column vector representing $|\theta\rangle$ and the row vector representing $\langle\psi|$. (The order of vectors is important!)

Solution/Hints $\rightarrow$

$$\hat{A} = |\theta\rangle\langle\psi| = \begin{pmatrix} \frac{i}{2} \\ 0 \\ \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{i}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} \frac{i}{2\sqrt{3}} & -\frac{i}{2\sqrt{3}} & \frac{1}{2\sqrt{3}} \\ 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{i}{2} \end{pmatrix}$$

- (e) **[SELF]** Is it clear why $A_{ij} = \langle \phi_i | \hat{A} | \phi_j \rangle$ are called the “matrix elements”?

Solution/Hints $\rightarrow$

Because these are exactly the elements of a matrix representation of operators.

- (f) **[2 pts.]** Using matrix-vector multiplication, find the state obtained by operating with $\hat{A}$ on the state $|W\rangle$.

Solution/Hints $\rightarrow$

$$\hat{A}|W\rangle = \begin{pmatrix} i/2\sqrt{3} & -i/2\sqrt{3} & 1/2\sqrt{3} \\ 0 & 0 & 0 \\ 1/2 & -1/2 & -i/2 \end{pmatrix} \begin{pmatrix} \gamma \\ \gamma \\ \gamma \end{pmatrix} = \begin{pmatrix} \gamma/2\sqrt{3} \\ 0 \\ -i\gamma/2 \end{pmatrix} = \begin{pmatrix} i/6 \\ 0 \\ 1/2\sqrt{3} \end{pmatrix}$$

4. A particle in a 1D harmonic oscillator: eigenstates.

For a single particle in a harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$, the time-independent Schroedinger equation has an infinite number of eigenvalue/eigenfunction pairs. The lowest three energy eigenvalues are $E_0 = \frac{1}{2}\hbar\omega$, $E_1 = \frac{3}{2}\hbar\omega$, $E_2 = \frac{5}{2}\hbar\omega$. The corresponding eigenfunctions are respectively

$$\begin{aligned}\phi_0(x) &= A_0 e^{-x^2/2\sigma^2}, & \phi_1(x) &= A_1 x e^{-x^2/2\sigma^2}, \\ \phi_2(x) &= A_2 \left(\frac{2x^2}{\sigma^2} - 1 \right) e^{-x^2/2\sigma^2}. & \text{Here } \sigma^2 &= \frac{\hbar}{m\omega}.\end{aligned}$$

- (a) [**1+1 pts.**] Sketch (by hand) plots of the three functions $\phi_n(x)$ for $n = 0$, $n = 1$, $n = 2$. You may use a computer program if you want to figure out how these functions look like, but please submit hand-drawn sketches.

By looking at the three plots and observing the pattern, guess and sketch the plots of the next two eigenstates, $\phi_3(x)$ and $\phi_4(x)$.

Solution/Hints →

As we increase n , the eigenfunctions have more and more 'nodes', i.e., zeros. The $n = 0$ case is the gaussian, which has no nodes. The next eigenfunctions have one, two, three, four,.... nodes.

- (b) [**2 pts.**] Look up (or calculate, or ask a computer algebra system or online integrator) the integrals

$$\int_{-\infty}^{+\infty} x^{2n} e^{-x^2} dx$$

for $n = 0$, $n = 1$, $n = 2$. Report the three results.

The first one ($n = 0$, integral over a Gaussian) is important enough that you might consider learning it for life.

Solution/Hints →

$$\begin{aligned}\int_{-\infty}^{+\infty} e^{-x^2} dx &= \sqrt{\pi} & \int_{-\infty}^{+\infty} x^2 e^{-x^2} dx &= \frac{\sqrt{\pi}}{2} \\ \int_{-\infty}^{+\infty} x^4 e^{-x^2} dx &= \frac{3\sqrt{\pi}}{4}\end{aligned}$$

- (c)
- [SELF]**
- Calculate the integrals

$$\int_{-\infty}^{+\infty} x^{2n} \exp \left[-\frac{x^2}{2\sigma^2} \right] dx$$

for $n = 0, n = 1, n = 2$, based on the integrals above.**Solution/Hints** $\rightarrow$

$$\left. \begin{aligned} \int_{-\infty}^{+\infty} \exp \left[-\frac{x^2}{2\sigma^2} \right] dx &= \sqrt{2\pi}\sigma \\ \int_{-\infty}^{+\infty} x^2 \exp \left[-\frac{x^2}{2\sigma^2} \right] dx &= \sqrt{2\pi}\sigma^3 \\ \int_{-\infty}^{+\infty} x^4 \exp \left[-\frac{x^2}{2\sigma^2} \right] dx &= 3\sqrt{2\pi}\sigma^5 \end{aligned} \right| \begin{aligned} \int_{-\infty}^{+\infty} \exp \left[-\frac{x^2}{\sigma^2} \right] dx &= \sqrt{\pi}\sigma \\ \int_{-\infty}^{+\infty} x^2 \exp \left[-\frac{x^2}{\sigma^2} \right] dx &= \frac{\sqrt{\pi}}{2}\sigma^3 \\ \int_{-\infty}^{+\infty} x^4 \exp \left[-\frac{x^2}{\sigma^2} \right] dx &= \frac{3\sqrt{\pi}}{4}\sigma^5 \end{aligned}$$

- (d)
- [SELF]**
- Use symmetry arguments to evaluate the integral

$$\int_{-\infty}^{+\infty} x^{2n+1} \exp \left[-\frac{x^2}{2\sigma^2} \right] dx$$

for any integer n . The integrand is an odd function.**Solution/Hints** $\rightarrow$

ZERO

- (e)
- [1+1+2 pts.]**
- Show that
- ϕ_0
- and
- ϕ_1
- are orthogonal to each other, as are
- ϕ_1
- and
- ϕ_2
- , and finally that
- ϕ_0
- and
- ϕ_2
- are orthogonal to each other.

Solution/Hints $\rightarrow$ $\phi_0(x)\phi_1(x)$ is an odd function of x . Hence $\int_{-\infty}^{\infty} dx \phi_0(x)\phi_1(x) = 0$. ϕ_1 and ϕ_2 are orthogonal for the same reason. ϕ_0 and ϕ_2 : needs more work

$$\begin{aligned} \int_{-\infty}^{\infty} dx \phi_0(x)\phi_2(x) &= A_0 A_2 \int_{-\infty}^{\infty} dx e^{-x^2/2\sigma^2} \left(\frac{2x^2}{\sigma^2} - 1 \right) e^{-x^2/2\sigma^2} \\ &= A_0 A_2 \left[\frac{2}{\sigma^2} \int_{-\infty}^{\infty} dx x^2 e^{-x^2/2\sigma^2} - \int_{-\infty}^{\infty} dx e^{-x^2/2\sigma^2} \right] \\ &= A_0 A_2 \left[\frac{2}{\sigma^2} \left(\frac{\sqrt{\pi}}{2} \sigma^3 \right) - \sqrt{\pi}\sigma \right] = 0 \end{aligned}$$

- (f) [5 pts.] Show in general that, if $|w_1\rangle$ and $|w_2\rangle$ are eigenstates of a hermitian operator corresponding to distinct eigenvalues, then they are orthogonal to each other. (This is an important result.)

Solution/Hints $\rightarrow$

Let the operator be $\hat{A}$, and the corresponding eigenvalues be a_1 and a_2 :

$$\hat{A}|w_1\rangle = a_1|w_1\rangle \quad \hat{A}|w_2\rangle = a_2|w_2\rangle$$

Since $\hat{A}$ is hermitian, $\hat{A} = \hat{A}^\dagger$. Also, a_1 and a_2 are real. So, the first equation gives the dual relation:

$$\langle w_1| \hat{A}^\dagger = a_1^* \langle w_1| \quad \implies \quad \langle w_1| \hat{A} = a_1 \langle w_1|$$

Applying $\langle w_1|$ on the second eigenvalue equation:

$$\langle w_1| \hat{A} |w_2\rangle = \langle w_1| a_2 |w_2\rangle \quad \implies \quad a_1 \langle w_1|w_2\rangle = a_2 \langle w_1|w_2\rangle$$

Since $a_1 \neq a_2$, this implies $\langle w_1|w_2\rangle = 0$, i.e., that the two kets are orthogonal to each other.

This is really a very important result; please make sure you can reproduce the proof. E.g., using the fact that the Hamiltonian is hermitian, prove that eigenstates corresponding to different eigen-energies are orthogonal to each other.

- (g) [1+2 pts.] Calculate A_0 and A_1 so that ϕ_0 and ϕ_1 are normalized.

Solution/Hints $\rightarrow$

$$\int_{-\infty}^{\infty} dx |\phi_0(x)|^2 = |A_0|^2 \int_{-\infty}^{\infty} dx \exp\left[-\frac{x^2}{\sigma^2}\right] dx = |A_0|^2 \sqrt{\pi}\sigma$$

Normalization requires the norm above to be unity. Taking A_0 to be real,

$$A_0 = \frac{1}{\sqrt{\sqrt{\pi}\sigma}} = \frac{1}{\pi^{1/4}\sigma^{1/2}}$$

A_1 is calculated similarly:

$$A_1 = \sqrt{\frac{2}{\sqrt{\pi}\sigma}} = \frac{\sqrt{2}}{\pi^{1/4}\sigma^{3/2}}$$

- (h) [**SELF**] Calculate A_2 so that ϕ_2 is normalized.

Solution/Hints →

Calculated similarly as above, but much more tedious. I believe the answer should be

$$A_2 = \sqrt{\frac{1}{\sqrt{\pi}2\sigma}} = \frac{1}{\pi^{1/4}(2\sigma)^{1/2}}$$

- (i) [**1 pt.**] Construct a solution of the TDSE, based on the three eigenfunctions of the time-independent SE.

Solution/Hints →

The solutions of the TDSE corresponding to the three time-independent wavefunctions are

$$\begin{aligned}\phi_0(x)e^{-E_0t/\hbar} &= \phi_0(x)e^{-\frac{1}{2}\omega t} \\ \phi_1(x)e^{-E_1t/\hbar} &= \phi_1(x)e^{-\frac{3}{2}\omega t} \\ \phi_2(x)e^{-E_2t/\hbar} &= \phi_2(x)e^{-\frac{5}{2}\omega t}\end{aligned}$$

The general solution is

$$B\phi_0(x)e^{-\frac{1}{2}\omega_0 t} + C\phi_1(x)e^{-\frac{3}{2}\omega_0 t} + D\phi_2(x)e^{-\frac{5}{2}\omega_0 t}$$

- (j) [**3 pts.**] At time $t = 0$, the particle in the harmonic oscillator finds itself in the state

$$\psi(x, 0) = \frac{1}{\sqrt{2}}\phi_0(x) - \frac{1}{2}\phi_1(x) + \frac{1}{2}\phi_2(x).$$

What is the wavefunction (state) $\psi(x, t)$ at a later time t ?

Solution/Hints →

Comparing with the general solution written above, we see that the given initial state fixes the constants B , C , D . Thus the wavefunction at arbitrary later time is given by

$$\psi(x, t) = \frac{1}{\sqrt{2}}\phi_0(x)e^{-\frac{1}{2}\omega t} - \frac{1}{2}\phi_1(x)e^{-\frac{3}{2}\omega t} + \frac{1}{2}\phi_2(x)e^{-\frac{5}{2}\omega t}$$

- (k) **[SELF]** If the $\phi_n(x)$ functions are normalized, then show that $\psi(x, 0)$ is normalized, and that $\psi(x, t)$ remains normalized at later times t .

Solution/Hints $\rightarrow$

This is an instructive exercise; I suggest trying it. You will also have to use the fact that the $\phi_n(x)$ functions are orthogonal to each other. You will not have to do any integrals.